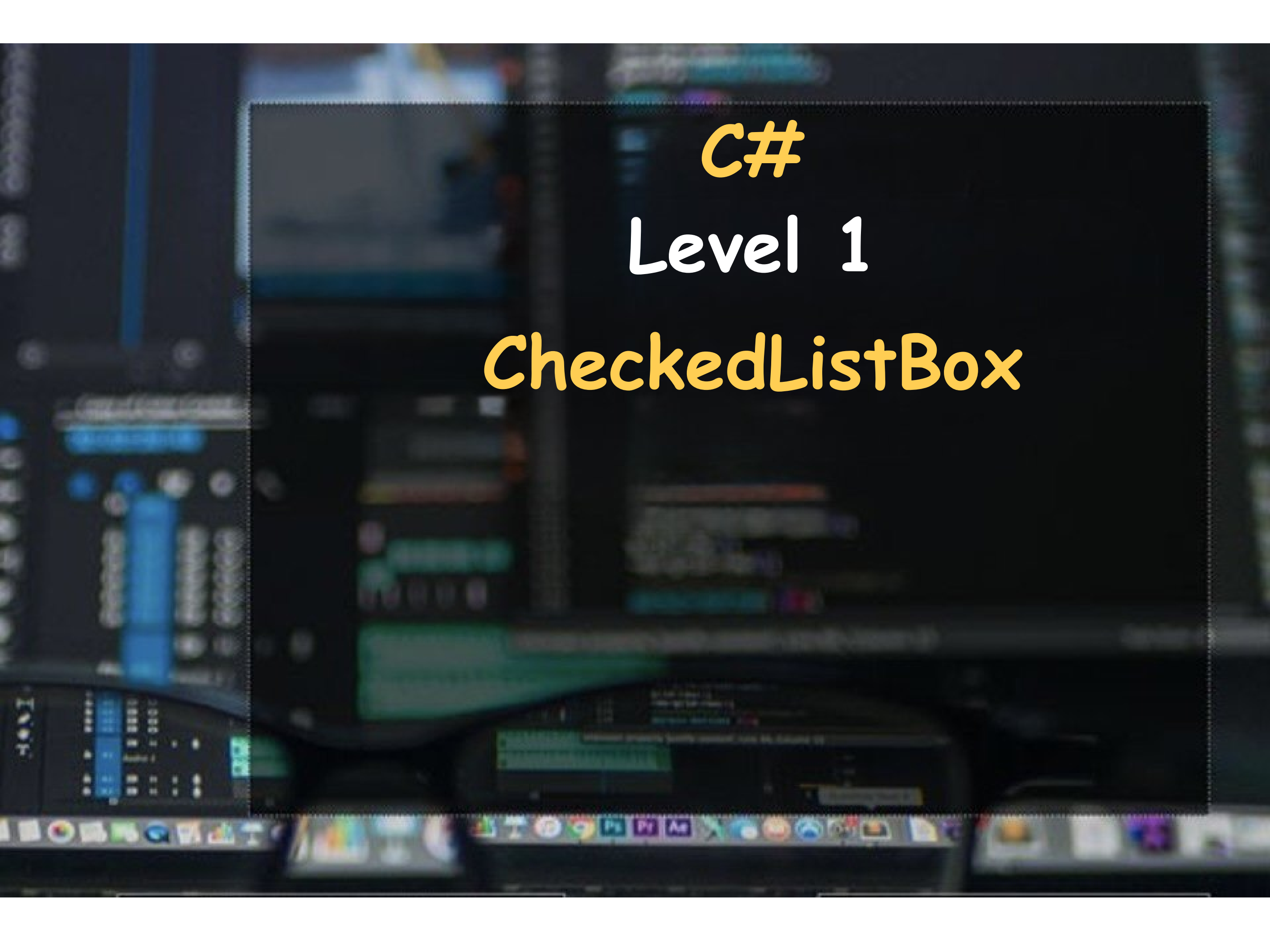




C#

Level 1

CheckedListBox



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Definition

CheckedListBox

 Windows Forms control that displays a list of items with a checkbox next to each item.

 Allows the user to check multiple items at the same time.

Explanation

Adding Items

-  Items are added using ``Items.Add()`.`
-  A ``for`` loop can be used to add multiple items.

Accessing Checked Items

-  ``CheckedItems`` contains the items that are currently checked.
-  ``CheckedIndices`` contains the indexes of checked items.

Checking Items Programmatically

-  ``SetItemChecked(index, true)`` checks an item.
-  ``SetItemChecked(index, false)`` unchecks an item.
-  ``GetItemChecked(index)`` returns whether an item is checked.

Procedure

1. Add items using `Items.Add()`.
2. Use `CheckedItems` to access checked items.
3. Display checked items using `MessageBox.Show()`.
4. Use `SetItemChecked()` to check or uncheck items.
5. Use `RemoveAt()` to remove an item by index.

💡 Example

```
for (int i = 1; i <= 5; i++)  
{  
    checkedListBox1.Items.Add("Item" + i);  
}
```

▣ This adds:

Item1, Item2, Item3, Item4, Item5

▣ To display checked items:

```
for (int i = 0; i < checkedListBox1.CheckedItems.Count; i++)  
{  
    MessageBox.Show(checkedListBox1.CheckedItems[i].ToString());  
}
```

➡ Result

▣ If Item1 and Item3 are checked:

▣ `MessageBox` displays `Item1`.

▣ `MessageBox` displays `Item3`.

▣ `SetItemChecked(i, true)` checks an item.

▣ `SetItemChecked(i, false)` unchecks an item.

Note

- ◆ `Items.Count` → total number of items.
- ◆ `CheckedItems.Count` → number of checked items.
- ◆ `CheckedIndices` → indexes of checked items.
- ◆ `RemoveAt(index)` → removes an item by index.
- ◆ `CheckOnClick` → controls whether clicking an item immediately changes its checked state.



Comparison

CheckedListBox

- ✓ Displays checkboxes.
- ✓ Supports multiple checked items.
- ✓ Uses `CheckedItems` and `CheckedIndices`.

ListBox

- ✓ Displays items without checkboxes.
- ✓ Uses `SelectedItems` for multiple selections.
- ✓ Does not have `CheckedItems`.

⊘ Warning

- ◆ Indexes start from `0`.
- ◆ `RemoveAt(2)` removes the third item.
- ◆ Removing an item changes the indexes of following items.
- ◆ Accessing an invalid index can cause `ArgumentOutOfRangeException`.

✦ Summary

- ✓ `CheckedListBox` is used for selecting multiple options with checkboxes.
- ✓ ``Items`` → all items.
- ✓ ``Items.Add()`` → adds an item.
- ✓ ``Items.Count`` → counts all items.
- ✓ ``CheckedItems`` → checked items.
- ✓ ``CheckedIndices`` → indexes of checked items.
- ✓ ``SetItemChecked()`` → checks or unchecks an item.
- ✓ ``GetItemChecked()`` → checks an item's state.
- ✓ ``RemoveAt()`` → removes an item by index.

Thank you 😊